Week 8 - Graded Assignment 8

The due date for submitting this assignment has passed.
Due on 2026-04-08, 23:59 IST.
You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-04-08, 08:45 IST

Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.

Multiple Select Questions (MSQ):
1 point
An inner product on R^3 R3 is defined as:

$$\langle .,. \rangle \langle .,. \rangle : R^3 \times R^3 \text{R3} \times \text{R3} \rightarrow R \rightarrow \text{R}$$

$$\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle \langle (\text{x1}, \text{x2}, \text{x3}), (\text{y1}, \text{y2}, \text{y3}) \rangle = x_1 y_1 + x_2 y_2 + x_3 y_3 \text{x1y1} + \text{x2y2} + \text{x3y3}.$$

Match the sets of vectors in column A with their properties of orthogonality or orthonormality in column B with respect to the above inner product.

Choose the set of correct options.

- [ ] a → → iv)
- [x]

a → → iii)

b → → iv)

b → → iii)

c → → ii)

c → → i)

d → → i)

d → → ii)

Yes, the answer is correct.

Score: 1

Accepted Answers:

a → → iii)
b → → iv)
c → → i)
d → → ii)

1 point

Choose the set of correct options.

Suppose $\beta = \{v_1, v_2, \ldots, v_n\}$ is an orthogonal basis of an inner product space V. If there exists some $v \in V$, such that $\langle v, v_i \rangle = 0$ for all $i = 1, 2, \ldots, n$, then $v = 0$.

There exists an orthonormal basis for R^n with the standard inner product.

If P_W denotes the linear transformation which projects the vectors of an inner product space V to a subspace W of V, then $range(P_W) \cap null\ space(P_W) = \{0\}$, where 00 denotes the zero vector of V.

$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ cannot represent a matrix corresponding to some projection.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Suppose $\beta = \{v_1, v_2, \ldots, v_n\}$ is an orthogonal basis of an inner product space V. If there exists some $v \in V$, such that $\langle v, v_i \rangle = 0$ for all $i = 1, 2, \ldots, n$, then $v = 0$.
There exists an orthonormal basis for R^n with the standard inner product.
If P_W denotes the linear transformation which projects the vectors of an inner product space V to a subspace W of V, then $range(P_W) \cap null\ space(P_W) = \{0\}$, where 00 denotes the zero vector of V.
$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ cannot represent a matrix corresponding to some projection.

Numerical Answer Type (NAT):

If A is an orthogonal matrix of order 5, then find nullity of the matrix A.

0

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 0
1 point
Let $v \in R^3$ v ∈ R3 be a vector such that $||v|| = 5$||v|| = 5. If uu is the vector obtained from vv after the anti- clock wise rotation of XY-plane with angle 70° 70∘ about the Z- axis, then find the length of the vector uu.

5

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 5
1 point
Let $v = (1, 2, 2)$v = (1, 2, 2) be a vector in R^3 R3. If (a, b, c)(a, b, c) is the vector obtained from vv after the anti- clock wise rotation of YZ-plane with angle 60° 60∘ about the X- axis, then find the value of $a + b + c$a + b + c.

3

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 3
1 point

Consider a vector space $M_{2\times 2}(R)$M2×2(R) and a norm on the vector space defined as $\|A\| = \max\{|a_{11}| + |a_{21}|, |a_{12}| + |a_{22}|\}$‖A‖ = max{|a11| + |a21|, |a12| + |a22|}, where $A = [\begin{matrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{matrix}] \in M_{2\times 2}(R)$A = [a11a21a12a22] ∈ M2×2(R).

Let $B = [\begin{matrix} x & \sqrt{2}\,x \\ -\sqrt{2}\,y & y \end{matrix}]$B = [x− 2 √ y 2 √ xy] be an orthogonal matrix i.e. $BB^T = B^T B = I$BBT = BTB = I and assume $x, y > 0$x, y > 0. Then find the norm of the matrix $C = [\begin{matrix} \sqrt{3}\,x & \sqrt{3}\,x \\ \sqrt{3}\,y & \sqrt{3}\,y \end{matrix}]$C = [3 √ x 3 √ y 3 √ x 3 √ y] (i.e., $||C||$||C||)

2

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 2
1 point

Let $V = R^2$ V = R2 be the inner product space with usual inner product and a linear transformation $T: R^2 \to R^2$ T : R2 → R2 defined as $T(x, y) = (\frac{a}{\sqrt{a^2+2^2}}x + \frac{3}{\sqrt{b^2+3^2}}y, \frac{2}{\sqrt{a^2+2^2}}x + \frac{b}{\sqrt{b^2+3^2}}y)$T (x, y) = (a2+22

√

ax + b2+32

√

3y, a2+22

√

2x + b2+32

√

by). If TT is an orthogonal linear transformation, then find the value of $3a + 2b$.3a + 2b.

0

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 0

1 point

Comprehension Type Question:

With a particular frame of reference (in R^3 R3), position of a target is given as the vector $(3, 4, 5)$ $(3, 4, 5)$. Three shooters S_1 S1, S_2 S2, and S_3 S3 are moving along the lines $x = y$x = y, $x = -y$ x = −y, and $x = 2y$x = 2y, on the XYXY -plane (i.e., $z = 0$z = 0) to shoot the target. Suppose that, there is another shooter S_4 S4, who is moving on the plane $x + y + z = 0$x + y + z = 0. Suppose all of them shoot the target so that the target is at the closest distance from the respective path or plane on which they are travelling. Answer questions 10,11 and 12 using the given information.

1 point

Choose the set of correct options.

- [ ] S_1 S1 will shoot the target from the point $(\frac{7}{2}, -\frac{7}{2}, 0)(27, -27, 0)$.
- [x] S_1 S1 will shoot the target from the point $(\frac{7}{2}, \frac{7}{2}, 0)(27, 27, 0)$.
- [ ] S_1 S1 will shoot the target from the point $(1, 1, 0)(1, 1, 0)$.
- [ ] S_2 S2 will shoot the target from the point $(-\frac{1}{2}, -\frac{1}{2}, 0)(-21, -21, 0)$.
- [ ] S_2 S2 will shoot the target from the point $(1, -1, 0)(1, -1, 0)$.
- [x] S_2 S2 will shoot the target from the point $(-\frac{1}{2}, \frac{1}{2}, 0)(-21, 21, 0)$.
- [x] S_3 S3 will shoot the target from the point $(4, 2, 0)(4, 2, 0)$.
- [ ] S_3 S3 will shoot the target from the point $(2, 1, 0)(2, 1, 0)$.
- [ ] S_3 S3 will shoot the target from the point $(0, 0, 0)(0, 0, 0)$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

S_1 S1 will shoot the target from the point $(\frac{7}{2}, \frac{7}{2}, 0)$(27, 27, 0).

S_2 S2 will shoot the target from the point $(-\frac{1}{2}, \frac{1}{2}, 0)$(−21, 21, 0).

S_3 S3 will shoot the target from the point (4, 2, 0)(4, 2, 0).

If (a, b, c)(a, b, c) is the point from which the shooter S_4 S4 will shoot the target, then find the value of $a + 2b + 3c$a + 2b + 3c.

2

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 2

1 point

Let d_i di be the distance of the target from the point where the shooter S_i Si shoots the target, for i = 1,2,3,4 and let dd be the minimum amongst the d_i di. Find the value of d^2 d2.

25.5

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 25.5

1 point